

2023

Bachelor in Business Administration

Course Title: Investment Analysis

Code No: **BNK204** Semester: **7th Semester**

Full Marks: 60 Pass Marks: 24 Time: 1 hour

1. The value-weighted index considers both market price per share and number of shares outstanding.

2. Blue chip stocks are large, well-established and well-known companies with long records of earnings and dividends.

3. A back end load is redemption fee incurred by mutual fund when you purchase share.

4. Optimal risky portfolio is the combination of a risky asset and risk-free asset.

5. Bond indenture is an agreement between the issuer and investor.

6. Price earnings ratio measures liquidity of the firm.

7. Federal Government issues municipal bond in Nepal.

Section "B" Short answer questions: [6 × 5 = 30]

1. Consider the following information: Stock price per share Rs 80, Margin requirement 50%, Interest rate on margin accounts 9%, Maintenance margin 30%. Ignoring transaction costs and taxes, assume that an investor takes a long position without using margin. Calculate the rate of return if the stock is sold for Rs 100 per share after one year.

2. Assume that an investor takes a long position using margin.

a. Calculate the stock price that will trigger margin call.

b. Calculate the rate of return if the stock is sold for Rs 100 per share after one year.

3. An investor has Rs 10,000 to invest. He is considering the following funds, all of which have a NAV of Rs 10 per share. Closed end fund 'A' is selling for a market price that equals to a NAV whereas closed end fund 'B' is selling at a discount of 20%. The broker charges a commission of 2% on the market price for each share purchased. Mutual fund 'C' has a front end load of 5.54% whereas mutual fund 'D' charges net 8.54% load.

a. How many shares does the investor end up in each case?

b. Which fund you should purchase?

4. Kamal purchased a bond selling at its face value of Rs 1,000. The bond has five years to maturity and a 10% coupon rate. The bond was called two years later for a price of Rs 1,160, after making its second annual interest payment. Kamal then reinvested the proceeds in a bond selling at its face value of Rs 1,000, with three years to maturity and a 7% coupon rate. What was Kamal's actual yield-to-maturity over the five-year period?

5. Alpha Bank's stock and Beta bank's stock have the following probability distributions of expected future returns:

Probability	0.10	0.20	0.40	0.20	0.10
Return (Alpha)	-10.0%	5.10%	25.35%	35.0%	40.0%
Return (Beta)	-20.0%	0.15%	30.40%	40.0%	40.0%

a. Calculate the expected rate of return for each bank's stock.

b. Calculate the standard deviation of expected returns for each bank's stock.

c. Calculate coefficient of variation and which stock would you prefer?

6. The market consensus is that Electronic Corporation has an ROE = 9%, has a beta of 1.25, and plans to maintain indefinitely its traditional plowback ratio of 2/3. This year's earnings were Rs 3 per share. The annual dividend was just paid. The consensus estimate of the coming year's market return is 14%, and T-bills currently offer a 6% return.

a. Find the price at which Electric stock should sell.

b. Calculate the P/E ratio.

c. Calculate the present value of growth opportunities.

7. Describe the role of stock market. Also highlight the current status of Nepalese stock market.

Section "C" Comprehensive answer questions: [2 × 10 = 20]

1. Consider the following information for two mutual funds:

	Fund A	Fund B
Expected return	8%	13%
Standard deviation	12%	20%
Covariance between fund A and B	72	
Correlation coefficient between fund A and B	0.30	

a. Find out the minimum variance portfolio.

b. Calculate the expected return of the minimum variance portfolio.

c. Calculate the standard deviation of the minimum variance portfolio.

d. What conclusion can you draw from the above calculations?

2. These four common stocks issued no additional shares and had no stock dividends or splits. Ignore cash dividend payments when computing the price index.

Stock	Total shares outstanding on both dates	Date (July 16, 2020)	Date (July 16, 2021)
A	60,000	Rs 1,000	Rs 600
B	50,000	Rs 1,000	Rs 2000
C	100,000	Rs 200	Rs 300
D	50,000	Rs 100	Rs 40

a. Calculate and interpret the value of the index as on July 16, 2021 assuming that the new four-stock index is (i) value-weighted, (ii) price-weighted, and (iii) equally-weighted.

b. Describe the uses of market index.

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Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Commercial paper is secured promissory note issued by Nepal Rastra Bank.
2. Treasury bill of Rs 10,000 can currently be purchased for Rs 9,600. The rates of return an investor earn 4.166 percent from holding until it matures in 65 days.
3. The maintenance margin is the maximum margin that must be maintained at all times in a margin account.
4. The value-weighted index considers both market price per share and number of shares outstanding.
5. Blue chip stocks are large, well-established and well-known companies with long records of earnings and dividends.
6. A back end load is redemption fee incurred by mutual fund when you purchase share.
7. Optimal risky portfolio is the combination of a risky asset and risk-free asset.
8. Bond indenture is an agreement between the issuer and investor.
9. Price earnings ratio measures liquidity of the firm.
10. Federal Government issues municipal bond in Nepal.
11. Consider the following information: Stock price per share Rs 80 Margin requirement 50% Interest rate on margin accounts 9% Maintenance margin 30% Ignoring transaction costs and taxes a. Assume that an investor takes a long position without using margin. Calculate the rate of return if the stock is sold for Rs 100 per share after one year. b. Assume that an investor takes a long position using margin. i. Calculate the stock price that will trigger margin call. ii. Calculate the rate of return if the stock is sold for Rs 100 per share after one year.
12. An investor has Rs 10,000 to invest. He is considering the following funds, all of which have a NAV of Rs 10 per share. Closed end fund 'A' is selling for a market price that equals to a NAV whereas closed end fund 'B' is selling at a discount of 20%. The broker charges a commission of 2% on the market price for each share purchased. Mutual fund 'C' has a front end whereas mutual fund 'D' charges net 8.54% load. a. How many shares does the investor end up in each case? b. Which fund you should purchase?
13. Kamal purchased a bond selling at its face value of Rs 1,000. The bond has five years to maturity and a 10% coupon rate. The bond was called two years later for a price of Rs 1,160, after making its second annual interest payment. Kamal then reinvested the proceeds in a bond selling at its face value of Rs 1,000, with three years to maturity and a 7% coupon rate. What was Kamal's actual yield-to-maturity over the five-year period?
14. Alpha Bank's stock and Beta bank's stock have the following probability distributions of expected future returns:

Probability	0.10	0.20	0.40	0.20	0.10
Return (Alpha)	-10.0%	5.10	25.35	35.0%	20.0%
Return (Beta)	-20.0%	0.15	30.40	40.0%	0.0%

 a. Calculate the expected rate of return for each bank's stock. b. Calculate the standard deviation of expected returns for each bank's stock. c. Calculate coefficient of variation and which stock would you prefer?
15. The market consensus is that Electronic Corporation has an ROE = 9%, has a beta of 1.25, and plans to maintain indefinitely its traditional plowback ratio of 2/3. This year's earnings were Rs 3 per share. The annual dividend was just paid. The consensus estimate of the coming year's market return is 14%, and T-bills currently offer a 6% return. a. Find the price at which Electric stock should sell. b. Calculate the P/E ratio. c. Calculate the present value of growth opportunities.
16. Describe the role of stock market. Also highlight the current status of Nepalese stock market.
17. Consider the following information for two mutual funds.

Fund	A	B
Expected return	8%	13%
Standard deviation	12%	20%
Covariance between fund A and B	72	
Correlation coefficient between fund A and B	0.30	

 a. Find out the minimum variance portfolio. b. Calculate the expected return of the minimum variance portfolio. c. Calculate the standard deviation of the minimum variance portfolio. d. What conclusion can you draw from the above calculations?

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- 18.** These four common stocks issued no additional shares and had no stock dividends or splits. Ignore cash dividend payments when computing the price index. Stock Total shares outstanding on both dates
- | Stock | Price (July 16, 2020) | Price (July 16, 2021) | Total Shares |
|-------|-----------------------|-----------------------|--------------|
| A | 60 | 100 | 1,000 |
| B | 50 | 100 | 2,000 |
| C | 100 | 200 | 300 |
| D | 50 | 100 | 40 |
- a. Calculate and interpret the value of the index as on July 16, 2021 assuming that the new four-stock index is (i) value-weighted, (ii) price-weighted, and (iii) equally-weighted.
 b. Describe the uses of market index. Sonnet 4.5CI